# Haocong(David) Guo


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## RESEARCH INTERESTS

The interaction between algebraic, combinatorial and geometric aspects of group theory.

## EDUCATION

2021 - 2025 | PhD in MATHEMATICS, **University of Bristol**
Supervisor: Matthew Tointon

2016 – 2020 | Master's in MATHEMATICS, **University of Warwick**
Thesis supervisor: Derek Holt
Thesis topic: The complexity of the compressed word problem for finitely generated free groups

## PAPERS

**In progress**

1. Residually finite groups with uniformly almost flat quotients (With Matthew Tointon)
2. The Conjugacy Ratio of abelian-by-cyclic groups

## TALKS GIVEN

Oct 2024 | Bristol Combinatorics Seminars, **University of Bristol**

Feb 2024 | Bristol Junior Algebra and Combinatorics Seminar, **University of Bristol**
Uniformly almost flat coset space in polycyclic groups

July 2023 | Postgraduate Group Theory Conference 2023, **Heriot-Watt University**
Uniformly almost flat coset space in polycyclic groups

May 2023 | Bristol Junior Geometry Seminar, **University of Bristol**
Uniformly almost flat coset space in polycyclic groups

Dec 2022 | Seminars on Groups and Graphs (online), **SUSTech**
Random Walks on Vertex-Transitive Graphs with Moderate Growth

Oct 2022 | Random walks and related random topics, **University of Göttingen**
Random Walks on Vertex-Transitive Graph with Moderate Growth

## OTHER CONFERENCES

July 2024 | Summer School on Symmetry and Randomness, **University of Bristol**

Apr 2024 | Young geometric group theory XII, **University of Bristol**

June 2023 | Probabilistic Group Theory, **University of Bristol**

Jan 2023 | LMS Regional Meeting, Geometric Group Theory, **University of Southampton**

Sept 2022 | Young Group Theorists Workshop, **SwissMAP Research Station**

July 2022 | Graduate Student Concentration Week on Metric Geometry, **Texas A & M University**

July 2022 | Postgraduate Group Theory Conference, **King's College London**

## TEACHING EXPERIENCE

**University of Bristol (2021-present):**

2024 - 2025 MATH10011 Analysis, Support Class Teacher
2024 - 2025 MATH33300 Group theory, Support Class Teacher
2023 - 2024 MATH10010 Proofs and Group Theory, Support Class Teacher
2022 - 2023 MATH10010 Linear algebra, Support Class Teacher
2022 - 2023 MATH33300 Group theory, Support Class Teacher

**Before 2021:**
2021 - 2022 Learning support tutor, Havering College, London
22017 - 2017 MA131 Analysis, Peer Tutor, University of Warwick

## VOLUNTEERING AND OUTREACH

July 2019 | Team Guide for IMO, **University of Bath**

Aug 2019 | Volunteer teacher and trainer, **Haryana, India**

## SERVICES

2025 | Postgraduate Group Theory Conference, **University of Bristol**, Co-organizer

2024 - present | Bristol Junior Algebra and Combinatorics Seminar, **University of Bristol**, Co-organizer